
DEPARTMENT OF MATHEMATICS

Mathematics Lab Assessment No. 2

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Question	An automated assembly line uses robotic grippers to assemble electronic control units. Historical maintenance records indicate that gripper failures occasionally occur due to wear, contamination, or pneumatic leakage. The maintenance department has collected the following information: • Probability that a gripper fails during a production shift = 0.04 • If a gripper is failing, the monitoring system generates a warning signal with probability 0.92 • If a gripper is operating normally, the monitoring system still generates a false warning with probability 0.05 Calculate: 1. Probability that a gripper is operating normally. 2. Probability that the monitoring system generates a warning. 3. Probability that a gripper is actually failing given a warning. 4. Probability of a false warning. 5. Reliability of the monitoring system.
Solution	i) Identify the parameters and mathematical concept $P(F)$ = Probability of failure = 0.04 $P(N)$ = Probability of normal operation = 0.96 $P(W/F)$ = Probability of warning given failure = 0.92 $P(W/N)$ = Probability of false warning = 0.05 Mathematical Concepts • Law of Total Probability • Bayes' Theorem ii) Solve analytically

Given,

$$P(F) = 0.04$$

$$P(W/F) = 0.92$$

$$P(W/N) = 0.05$$

- 1) Probability that gripper is operating normally is

$$P(N) = 1 - P(F)$$

$$P(N) = 1 - 0.04$$

$$P(N) = 0.96$$

96% of production shifts occur without gripper failure.

- 2) Probability that the monitoring system generates a warning

$$P(W) = P(W/F)P(F) + P(W/N)P(N)$$

$$P(W) = (0.92)(0.04) + (0.05)(0.96)$$

$$P(W) = 0.0368 + 0.048$$

$$P(W) = 0.0848$$

8.48% of production shifts generate a warning.

- 3) Probability that a gripper is actually failing given a warning

$$P(F/W) = \frac{P(W/F)P(F)}{P(W)}$$

$$P(F/W) = \frac{0.92 \times 0.04}{0.0848}$$

$$P(F/W) = 0.43396$$

Only about 43.39% of warning signals correspond to actual failures.

- 4) Probability of a false warning

$$P(\text{False Warning}) = P(W/N)P(N)$$

$$= (0.05)(0.96)$$

$$= 0.048$$

$$4.8\%$$

- 5) Reliability of the monitoring system

$$\text{Reliability} = P(F/W)$$

$$= 43.39\%$$

This means less than half of all warnings indicate a true failure.

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# Given Data
P_F = 0.04      # Probability of failure
P_W_F = 0.92    # Warning given failure
P_W_N = 0.05    # False warning probability

# Step 1: Probability of normal operation
P_N = 1 - P_F

# Step 2: Probability of warning
P_W = (P_W_F * P_F) + (P_W_N * P_N)

# Step 3: Bayes Theorem
P_F_W = (P_W_F * P_F) / P_W

# Step 4: False warning probability
P_false_warning = P_W_N * P_N

# Step 5: Reliability of monitoring system
reliability = P_F_W * 100

# Display Results
print("Probability of Normal Operation P(N):", round(P_N, 4))
print("Probability of Warning P(W):", round(P_W, 4))
print("Probability of Failure Given Warning P(F|W):", round(P_F_W, 4))
print("Probability of False Warning:", round(P_false_warning, 4))
print("Monitoring System Reliability:", round(reliability, 2), "%")
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iv) Results and analysis from the graph

Probability of Normal Operation $P(N)$: 0.96
Probability of Warning $P(W)$: 0.0848
Probability of Failure Given Warning $P(F|W)$: 0.434
Probability of False Warning: 0.048
Monitoring System Reliability: 43.4 %

- The reliability of the monitoring system is 43.39%, which means that when a warning signal is generated, there is only a 43.39% chance that the gripper is actually failing.
- Since only 43.39% of warnings correspond to actual failures, immediate maintenance after every warning could lead to unnecessary inspections.
- Although the detector is highly sensitive (92%), failures are rare (4%), and false warnings still occur 5% of the time. Therefore many warnings originate from normal conditions.

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	• Low failure rates increase the proportion of false positives among all warnings, reducing confidence in warning signals. • The probability that a warning represents a real failure increases. The monitoring system becomes more trustworthy. • Engineers should not rely only on warning signals they should combine warning signals with: ▪ Additional diagnostics ▪ Historical maintenance data ▪ Sensor trends ▪ Physical inspection ▪ before scheduling maintenance.